Referee report

Can Diminishing Returns Restrain an AI Race? Commercial Incentives, Delegation, and Resource Control

Simulated economics-theory review

21 October 2026

Recommendation and scope

I recommend rejection in the paper's present form at the requested serious general-interest economics journal standard. The complete replication succeeds, and the named mathematical results are substantially correct. The central obstacle is that the paper has not yet converted an economically sensible distinction into a sufficiently substantial new characterization. Its principal constructive results mostly restate optimality conditions, exclusion restrictions, and feasible-set comparisons. The numerical exercise adds mechanisms, but its strongest stopping example is largely determined by research indivisibility and assumed financing. There are also specific computational claims that require correction.

This assessment concerns the frozen draft identified as commit:

5a1d9a436f29e3a69ef434aa1966c0bcf87b3b7e

I read the complete 24-page manuscript, including all eight propositions, both theorems, the lemma, proofs, appendices, tables, and figures. I inspected the actual model, estimation, experiment, checking, and table-generation code. The Oracle bundle and repository snapshot contain substantively matching manuscript/code material; some bundled code omits blank lines.

I accessed GitHub's public API for that exact commit. All **53 tracked files** in its tree match the uploaded snapshot, allowing for CRLF versus LF line endings in 21 text files; 32 match byte for byte. The attached PDF is byte-for-byte identical to the snapshot's rendered PDF. I did not review subsequent revisions.

I ran the unmodified **python replication/run.py** in a separate copy under Python 3.12.14 with the pinned packages: numpy 2.5.2, scipy 1.18.1, pandas 3.0.5, matplotlib 3.11.1, pyarrow 23.0.1, and PyMuPDF 1.28.2. The run verified all five frozen data hashes, passed its checks, regenerated estimates/scenarios/tables, compiled with pdflatex and bibtex, and rendered all 24 pages. Runtime was 56.3 seconds; the build reported zero overfull boxes. I also inspected page renders of the supplied manuscript.

The largest absolute differences from supplied results were 2.274×10^{-13} in model results and 7.955×10^{-9} in empirical results. All generated manuscript tables and macros match after newline normalization. I found no substantive numerical transcription or replication discrepancy. Separately identified model checks and sensitivity calculations below were conducted for this report; they are not author-supplied results.

The original cached reference-paper PDFs are not in the snapshot. I consulted separately accessible primary sources: the published Jones paper, the April 24 Growiec–Prettner version, the May 2026 BFI version of Bueno de Mesquita–Dziuda–Polborn, Cunningham et al.'s arXiv material, and source documentation for Arena and ECI. The inaccessible original cache was not needed to run the frozen replication. I did not reproduce the source providers' benchmark construction or Fradkin's separate analysis.

The report distinguishes **essential repairs** to validity or interpretation from **extensions**. Successful reproduction establishes what this implementation computes. It does not identify real-world control failure, human subsistence thresholds, or extinction probabilities. The manuscript generally makes that distinction correctly.

1. Mathematical correctness

1.1 Verdict on every named result

Result	Assessment
Proposition 1: Counterexamples	Correct. The safe optimum payoff is 16.646647, while the dangerous optimum at $a = 2.301585$ gives 16.678415. Mortality destroys both commercial benefits and the value of life. The linear and increasing-benefit examples also work.
Proposition 2: Investment-envelope threshold	Correct under the stated differentiability, strict concavity, coercivity, and $\bar{a} > 1$ assumptions. The lower-boundary solution $a^* = 1$ is covered. These restrictions must hold separately at every compared curvature parameter.
Lemma 1: Capability-scale dependence	Correct. Both the transformed second derivative and the invariance of fully transformed choices/threshold orderings are valid.
Theorem 1: Existence	Correct for the stated finite game. Backward induction after autonomous optimization, followed by Nash existence in the government game, establishes time-dependent Markov subgame-perfect equilibrium. It says nothing about convergence to a continuous-state/action model.
Theorem 2: No-delegation certificate	Correct as a subgame-perfect/Markov-perfect characterization of the specified restricted equilibrium. It certifies existence of one no-delegation equilibrium, not safety of every equilibrium or endogenous policy choice.
Proposition 3: Local threshold	Correct with fixed continuation values and the stated independence restriction. The $H > 0$, $H = 0$, and $H < 0$ cases are handled appropriately.
Proposition 4: Prisoner's dilemma	Correct. The dominance bounds and interval length $qB/(1-q)^2$ are right. The payoffs 10.25, 10, 9.215, and 9, and extinction probability 0.19, check out.
Proposition 5: Conditional irrelevance	Correct as a parameter-exclusion result, holding all autonomous and human-continuation primitives and selection fixed. It does not establish equilibrium irrelevance of commercial returns.
Proposition 6: Acquisition and subsistence	Correct for the two-action comparison, common current reward, and $J_1 > 0$. With more actions, pairwise inequalities alone do not select the global optimum; the text recognizes this.

Result	Assessment
Proposition 7: Conditional/unconditional risk	Correct application of total probability to first-loss dates and states. Pure strategies can generate positive risk, and the extinction probability cannot exceed first-loss probability under the exclusive-channel assumption.
Proposition 8: Welfare comparisons	Correct feasible-set dominance under the specified objectives and continuation selection. Neither lower extinction risk nor a Pareto improvement follows.

Thus, the main criticisms should not be framed as discoveries that the elementary proofs are false.

1.2 Required precision and boundary assumptions

Severity: minor formal repairs, with one useful clarification.

In Sections 3–5, state explicitly that controlled and autonomous capability is positive, preferably $a \geq 1$ for the specialization. Otherwise the contest share is undefined at $(0, 0)$ and the commercial formula or negative powers can be undefined. The unbounded-progress example in Section 5.2 also needs $a_0 > 0$ when $\psi > 0$. State $\bar{K} > k_0$ and initial human resources at least $\underline{H}$ so that equation (14) and initial survival are well defined. The numerical parameters already satisfy these conditions.

In Theorem 2, replace unqualified "equilibrium" with "Markov-perfect/subgame-perfect equilibrium of the development game." An inequality at every off-path state is necessary for sequential optimality, but need not be necessary for an initial-state Nash equilibrium alone.

For outcome-dependent failure in Section 4.2, the phrase "use the corresponding conditional expectations" is too compressed. Let X denote the delegated research outcome and $H(X) = V(X) - Z(X)$. A transparent general identity is

$$Q_d - Q_s = N - \beta\, \mathbb{E}[q(X)H(X)].$$

If $q = \mathbb{E}[q(X)]$, this is

$$N - \beta q\, \mathbb{E}[H(X)] - \beta\, \mathrm{Cov}(q(X), H(X)).$$

Equation (11) follows when the appropriate independence holds. Conditioning alternatively on success and failure is possible, but the definitions and the interpretation of N must change consistently. This is a clarification, not a failure of Proposition 3 under its maintained assumptions.

2. The main contribution falls short of the stated ambition

Severity: central editorial concern. Locations: Introduction; Proposition 2; Theorem 2; Propositions 3 and 5; Sections 4–5.

The rejection of the unrestricted conjecture is sound, but it is an insufficient target for a substantial theory paper. A concave benefit function can have marginal benefit above marginal cost at any given finite threshold. Conversely, a convex cost function can dominate a linear or convex benefit. The examples establish this elementary point effectively.

The constructive results currently add less than the introduction suggests. Proposition 2 is the sign of the derivative at a threshold under assumed global concavity. Theorem 2 is the one-shot-deviation principle evaluated using already solved continuation values. Proposition 3 rearranges the difference between two expected payoffs. Proposition 5 removes commercial curvature from every relevant primitive and concludes that the autonomous solution is invariant to it. Each is correct, but none yet identifies a substantial new restriction in economically interpretable primitives.

In particular, the certificate does not yet make the stopping question easier to answer: evaluating it requires the entire restricted dynamic game and autonomous continuation. Moreover, the numerical code does not actually evaluate the certificate that serves as the theoretical centerpiece. It solves unrestricted games and classifies selected outcomes.

The existing literature is appropriately cited; I do not attribute a claim of originality for Bellman equations or finite-game existence to the author. The comparison nevertheless makes the remaining gap clear. Jones characterizes growth-versus-survival trade-offs; Growiec and Prettner explicitly analyze misaligned resource allocation and life-sustaining consumption; Bueno de Mesquita, Dziuda, and Polborn obtain market-structure and resource comparative statics in a speed–safety race; Cunningham et al. distinguish research and broader economic feedbacks. Adding these elements to one finite computational game needs to generate a sharper economic result than their coexistence. See the primary sources listed at the end of this report.

Highest-value revision now: solve the restricted no-delegation game and report

$$\Delta_{\max} = \max_{i,x,h,b_i:\, d_i=1} \left\{ \mathbb{E}_{\pi^0_{-i}} Q^0_{i,h}(x, b_i, b_{-i}) - V^0_{i,h}(x) \right\},$$

over feasible delegated actions, with the maximum over an empty set treated explicitly. Identify binding states, times, financing constraints, and the N and H components. Then derive economically informative sufficient conditions, or demonstrate which primitive restrictions consistently govern this margin. Distinguish an equilibrium-existence certificate from a guarantee applying to every equilibrium.

A stronger analytical direction is partial enforceability of a financing/resource constraint after delegation. Proposition 5 currently identifies the endpoint at which the constraint disappears entirely. A result about when an enforceable residual constraint is sufficient, and how its relaxation changes pre-delegation incentives, would address the paper's distinctive question more directly. This is a direction for genuine research, not a result that can be promised from the present assumptions.

3. The ceiling check can certify artificial stopping

Severity: essential numerical-validation repair. Locations: model.solve_verified_domain; selected-path boundary recursions in model.solve and model.autonomous; Section 8.1 and Appendix C.

The program expands ceilings only if capability clipping has positive probability on the selected equilibrium/autonomous path from the initial state. This does not establish that the ceiling is irrelevant to optimization. Truncation can suppress a profitable deviation, making holding optimal in the truncated game; the selected holding path then never clips.

I verified the following counterexample using the actual supplied routine. Start from the reference parameters, set $T = 1$, $\tau_H = 1$, $\sigma = 1.3$, positional prize zero, $\eta = 2.1$, research cost coefficient 0.75,

cost exponent zero, outside finance one, launch threshold 100, and $q_0 = 0$. Use controlled mesh 0.5, autonomous ceiling 10, and autonomous mesh 0.5. Change only the controlled ceiling.

Controlled ceiling	Selected initial actions	Total research cycles	Developer 1 value	Reported clipping probability
2	Hold / hold	0	9.800000	0
4	Supervised / supervised	2	9.948486	0

Both runs report zero domain expansions. From $a = 1$, supervised research targets

$$1 + \frac{2.1}{1 + 1/20} = 3.$$

The truncated private research gain is

$$0.96R_{1.3}(2) - 0.75 = -0.149208,$$

while the untruncated gain is

$$0.96R_{1.3}(3) - 0.75 = 0.148486.$$

This is artificial stopping under diminishing commercial marginal benefits, accepted by the current "verified domain" routine.

The supplied default sensitivity provides useful contrary evidence about magnitude in that particular case: doubling the ceilings changes the 25-year open/open extinction probability only from 0.0435854170 to 0.0435846376. This is very small. It does not repair the general validation criterion or certify every other scenario and policy pair.

Essential repair: construct horizon-specific domains containing states reachable under all feasible action histories from the modeled initial states, including deviations and all autonomous alternatives. The finite horizon and finite menus make such bounds feasible. Alternatively, provide valid continuation-error bounds and show that those errors cannot change best responses or relevant selections. Mandatory domain expansion and action/payoff stability checks for every reported specification are a useful interim sensitivity, but should not be described as a proof without an error bound.

Keep the present clipping probability as a selected-trajectory diagnostic. Do not use it alone as evidence of economically unconstrained stopping. Unilateral-deviation checks inside the same truncated game cannot detect this problem.

4. Equilibrium computation, interpolation, and horizon comparisons

4.1 The actual mixed-support search differs from the description

Severity: essential algorithm-description repair; no invalid baseline equilibrium found. Locations: model.equilibria and model.solve; Section 8.1.

The manuscript says no-pure-equilibrium games use linear feasibility programs to enumerate supports. In fact, model.solve calls equilibria with fast=True. That branch searches nonsingular equal-sized supports, starting with size two, and returns after the first support size yielding candidates. Larger, unequal, and degenerate supports are then skipped. LP enumeration is only a fallback.

The deviation check establishes approximate Nash validity of the returned profile. It does not establish the stated breadth of the search or minimum risk over the equilibrium correspondence.

For an explicit counterexample, take

$$A = \begin{pmatrix} 1 & 0 & -1 \\ -1 & -5 & 2 \\ -2 & -5 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} -3 & 0 & 3 \\ 2 & -3 & 1 \\ 2 & 1 & -2 \end{pmatrix}.$$

There is no pure equilibrium. The fast routine returns two 2×2-support equilibria; full enumeration also returns

$$p = (19, 6, 27)/52, \qquad q = (5, 1, 5)/11.$$

Assign risk one to every action pair except row three/column one, where risk is zero. Both fast-returned equilibria have risk one. The omitted equilibrium has risk $437/572 \approx 0.763986$.

I also checked materiality in the actual reference: replacing the fast search with full LP enumeration in all **26 no-pure state games** of the 25-year open/open specification leaves its reported outcomes exactly unchanged. The concern is general algorithm scope and the prose describing it, not a demonstrated reversal of the reference result.

Repair: enumerate all support pairs in these small stage games, or state precisely that the procedure selects a verified representative from a restricted search. Even one representative per feasible support does not exhaust a degenerate equilibrium face. The payoff-equivalent policy mixtures in the bridge specification illustrate why a finite list of representatives is not the whole equilibrium set.

4.2 Research outcomes differ across the control-failure branches

Severity: essential model/code reconciliation. Locations: equation (9), Section 3.1, Section 8.1; model.solve.

The text has research first produce x', followed by failure independent of x'. In the implementation, controlled continuation uses the lottery from projection onto the controlled grid. Failure continuation instead recomputes the continuous target and projects it directly onto the finer autonomous grid.

These are different numerical lotteries. They can define a legitimate regime-specific approximation, but they do not literally implement one shared realized x' followed by the stated mapping $M_j(x', b)$.

At reference capability $a = 2.5$, delegated research targets 3.782399916. With eleven autonomous periods remaining, direct autonomous-grid projection gives conditional extinction probability 0.800876825. Realizing the controlled-grid lottery between 3.5 and 4.0 first, then mapping that state into autonomy, gives 0.686276607. This is a substantial difference at that particular feasible state/action; I have not established that it lies on the reported selected path.

The aggregate reference is much less sensitive. A targeted common-outcome implementation changes 25-year open/open extinction probability from **4.35854% to 4.35529%**, and welfare from 774.8968 to 774.8916. Thus the mismatch is real, while its tested headline effect is small.

Repair: either draw/project research once and use that realized state in every branch, or formally define branch-specific projection kernels after a common unprojected target. Re-run the affected results under the intended specification. State clearly that these numerical lotteries are not estimated technological uncertainty.

4.3 Planning horizons and reporting horizons are different objects

Severity: essential interpretive clarification. Locations: Section 8.2; experiments.run; terminal conditions in model.solve and model.autonomous.

The ten-year and twenty-five-year cases re-solve both developer and autonomous choices under different horizons. They are not cumulative risks at different dates under one fixed equilibrium. The comparison combines additional time with changed incentives.

Moreover, autonomous terminal reward $\tau_A a$ gives no separate value to terminal resource holdings, while the human terminal rule freezes survival and technology. These approximations affect acquisition timing and delegation incentives. Varying both terminal coefficients together, as in the supplied sensitivities, does not isolate their separate effects.

Repair: solve a long-horizon economy once and report its cumulative first-loss and extinction distributions at several dates. Label existing rows as comparisons across planning horizons. Vary human and autonomous terminal values separately. An infinite-horizon solution is an extension; an infinite-horizon safety interpretation is unsupported by the present finite-horizon results.

5. The empirical-bridge halt depends on financing and indivisible research

Severity: essential economic-interpretation repair. Locations: Sections 7.3 and 8.1–8.2; equation (26); model.solve feasibility; experiments.run.

The bridge uses $\eta = 5.4564251$ and $\gamma = 1.8538057$. Initially $a = 1 < a_L = 2$, so delegation is unavailable. One supervised action targets

$$a' = 1 + \frac{5.4564251}{1.05} = 6.1965953.$$

At this target, the next unit research project costs

$$0.25(a')^{1.8538057} = 7.3525507,$$

while available financing is

$$0.65 + R_{1.3}(a') = 2.0547749.$$

The actual neighboring grid states, 6 and 6.5, also make another unit project unaffordable. Consequently every policy pair yields exactly two total supervised cycles, one per developer, and zero delegated cycles.

This is an informative mechanism, but its interpretation needs to be explicit. It combines a launch threshold, a very large first jump, the maintained cost bridge, fixed per-period financing without accumulated cash, and an indivisible unit research project. The code excludes subsequent research before comparing the profitability of those unavailable actions. The outcome does not establish that observed commercial curvature makes feasible dangerous research unattractive.

The supplied "ready to delegate" sensitivity sets $a_L = 1$ and produces two delegated cycles and loss-of-control probability 0.81036%. Halving the cost slope also produces positive delegation. These changes reinforce the role of the maintained eligibility and feasibility assumptions.

5.1 Small research inputs change exact stopping

The research menu is economically consequential. Suppose supervised effort can take arbitrarily small positive values, with no fixed cost. In the final development period, take the rival not to delegate, positive terminal value, positive infrastructure, and positive affordable expenditure. Write

$$A(a) = \eta \frac{a^{\phi-\psi}}{1 + a/B_R} > 0, \qquad a'(e) = a + A(a)e^{\alpha}.$$

Let $B_i(a, a'_j)$ denote terminal gross private flow. Since $R'_\sigma(a) > 0$ and the own-capability contest derivative is nonnegative,

$$\Delta Q_i(e) = \beta \tau_H A(a)\, \mathbb{E}[\partial_a B_i(a, a'_j)] e^{\alpha} - c_0 a^{\gamma} e + o(e^{\alpha}).$$

For the reference $\alpha = 0.6 < 1$, the positive $e^{0.6}$ term dominates the linear cost as e approaches zero. Holding is therefore not optimal in this clean last-period comparison. High research costs make the preferred input small; they do not force it exactly to zero.

This does not invalidate a proposition about the specified finite-action game. It shows that exact cessation requires a minimum project size, a fixed cost, satiation, no budget, or another explicit restriction in a smooth counterpart. Earlier-period continuation values can be nonmonotone because of rival responses, so this argument should not be overextended to every date.

Essential revision: distinguish infeasible research, feasible but unprofitable research, controlled innovation, and no delegation. Add smaller research actions, or justify a minimum project size as an economic primitive. Halving the capability-state mesh alone does not address this action indivisibility.

5.2 The autonomous effort menu adds an implicit acquisition cost

With $y = 1$, $k = 4$, and $\kappa_y = 0.5$, the budget permits research $e = 3.5$, but the menu $\{0, k/2, k\}$ permits at most $e = 2$. Relative to the maximum no-acquisition input, the menu therefore imposes a two-unit research sacrifice, although the explicit resource expenditure is only half a unit.

Including the affordable boundary $e = k - \kappa_y y$ is a cheap diagnostic. My targeted reference check changes extinction probability only from 4.35854% to 4.35756%. This supports local numerical stability while confirming that the menu is a substantive technological assumption.

Also separate the parameters in model.solve: the controlled safety infrastructure requirement uses seizure_cost, the autonomous acquisition-cost parameter. It coincides with the stated fixed 0.5 coefficient at baseline, but an acquisition-cost sensitivity would silently change safety feasibility too. This is a definite small code repair.

6. Timing and welfare accounting: mostly consistent, but the losses need decomposition

Severity: essential interpretation; no general mortality-accounting bug found. Locations: equations (6), (8), (9), and (14)–(19); model.solve and model.autonomous.

The implementation correctly pays current flows and costs before transition, discounts private and social values, leaves probability recursions undiscounted, and removes human continuation after extinction. The simultaneous-failure weights implement the stated fair selection and correlation mixture. Containment gives the system zero continuation and humans positive safe continuation. Once humans are extinct, containment is disabled; there is no inadvertent revival or payment to dead owners.

Three assumptions nevertheless materially shape the economic meaning:

First, control failure destroys the commercial sector's extra surplus permanently, even when containment succeeds. This is explicitly stated in Section 5.1 and implemented. Consequently policy can improve welfare by preventing permanent economic loss independently of mortality. Report surviving task surplus, research/safety costs, nonfatal resource losses, and mortality losses separately. A sensitivity allowing recovery of some post-containment commercial/task surplus is feasible now and would reveal how much the policy ranking rests on irreversible sector collapse.

Second, "value of life" is multiplied by human output while humans are still alive. The private continuation is $\ell_i L_h$, and the living-state flow in L_h is $g(k)$. Thus resource diversion reduces this life-value component before the subsistence threshold is crossed. Either interpret it as composite utility from life and consumption, or separate survival utility from material consumption. Merely increasing a parameter called private_life currently increases aversion to both kinds of loss.

Third, k behaves as persistent productive capacity. The system spends resource services satisfying $e + \kappa_y y \leq k$, yet next period has $k + y$; research does not deplete the stock. This is coherent for capital yielding a fresh service flow. It needs that interpretation, rather than an unqualified reference to consumable resources. Depreciation, replenishment, or alternative resource technologies are extensions, not required repairs to the finite optimization proof.

The numerical policy results should be interpreted as welfare comparisons under these particular utility and resource assumptions. They do not isolate a general willingness to pay for extinction-risk reduction.

7. The preference exercise is reproducible, but its inferential and commercial content is limited

7.1 Matching and attrition define a narrow population

Severity: substantial empirical limitation; attainable reporting repair. Locations: Section 7.1; Table 1; empirical.run; model_crosswalk.csv.

The explicit crosswalk, separation of nine qualified aliases, and first-round restriction are correctly implemented. The source defines evaluation order as vote order within a continuous evaluation session, so retaining the first observation addresses repeated votes in that session. It does not identify repeat users across sessions. Arena also documents changes in UI and task composition, which matter for generalization.

The sample flow is:

Selection	Comparisons
Raw frozen observations	135,634
Evaluation order one	108,315
Principal matches, first round, all outcomes	37,799
Ties excluded from those matches	5,847
Both-bad observations excluded	4,481
Final principal decisive sample	27,471

The final sample is 20.25% of raw observations and covers 32 models. This is a defensible descriptive estimand conditional on decisive comparisons within matched models, not a representative sample of AI demand.

Qualified aliases raise the sample to 52,764 observations over 41 models and change the quadratic estimate from -0.10057 to -0.01589. That is consequential composition/matching sensitivity. Add the attrition table and model-level comparison counts; distinguish changes in the estimand from increased precision. Identical model names alone do not establish comparable reasoning or inference budgets.

7.2 Pair clustering is implemented correctly, but model-level dependence matters

Severity: substantial evidential limitation; essential sensitivity if retaining inferential language. Locations: equation (23); empirical.fit_logit.

The published unordered-pair covariance is correctly computed for its maintained dependence structure. The manuscript also explicitly excludes common model shocks and capability-score uncertainty. The concern is the evidential weight placed on 27,471 comparisons when the capability regressor takes only 32 model-specific values.

As a sensitivity, I recalculated a dyadic score sandwich permitting dependence whenever observations share a model. If S_m sums score contributions incident to model m, and S_{mn} sums within unordered pair $\{m, n\}$, its meat is

$$\sum_m S_m S_m' - \sum_{\{m,n\}} S_{mn} S_{mn}'.$$

Using the actual frozen observations, without an extra finite-model correction, gives:

Parameter	Published pair-cluster SE	Illustrative dyadic SE
Linear ECI coefficient	0.05309	0.14454
Quadratic ECI coefficient	0.05706	0.15836

The quadratic normal interval changes from approximately $[-0.212, 0.011]$ to $[-0.411, 0.210]$. This is a change in dependence assumptions, not proof that the published conditional sandwich is arithmetically wrong, nor a claim that the illustrative alternative is definitive. It shows that an attainable robustness check further weakens the evidence.

State the inferential population, add model-node/dyadic sensitivity, and keep published-score uncertainty distinct. Model fixed effects cannot simply be added to this regression because model-level ECI regressors would be collinear with them.

7.3 Negative curvature is not uniformly positive diminishing marginal value

Severity: important interpretation. Locations: Table 1 and common-sample discussion.

The common sample gives GPQA curvature $+0.05029$ and ECI curvature -0.33908. More than scale dependence is involved: the measures can change model orderings, so they are not necessarily a monotone re-labeling covered by Lemma 1.

In the common-sample ECI quadratic, fitted latent utility turns downward above

$$E^* = 140 - \frac{5\widehat{\beta}_1}{\widehat{\beta}_2} = 140.3931,$$

inside its observed range 127.07–146.24. This is not positive but diminishing marginal value throughout the support. The principal-sample turning point is outside its observed range, so the specific criticism should be confined to the common-sample fit.

Show fitted levels and slopes over actual support, and consider a flexible or monotonicity-constrained descriptive comparison. Equation (24), its matched-support restriction, and the plotted coefficient-based pointwise bands are correctly implemented.

7.4 No causal or monetary bridge is identified

The paper appropriately avoids claiming willingness to pay, profits, or a monetary estimate of σ. Preserve that distinction. Capability is a model-level attribute bundled with architecture, post-training, style, speed, reasoning expenditure, and release timing. Even randomized assignment of responses would not independently randomize those attributes. Conditioning on a decisive vote adds another selection condition.

Without prices, quantities, an outside option, and cost information, latent choice utility cannot identify monetary returns. Coding prompts do not measure autonomous research productivity. These limitations are not cured by additional controls alone. The empirical section therefore does not test the model's central commercial stopping condition.

8. The cost and capability bridges are assumptions attached to verified moments

Severity: central to the quantitative contribution. Locations: Sections 7.2–7.3; equation (25); empirical.run; experiments.run.

The cost and progress calculations are correct. The 18 retained training runs give log-cost slope 1.853806 with HC1 SE 0.138364; adding publication year gives 1.861096 with SE 0.283953. The retrospectively scored frontier rises from 150.59 to 168.83, giving 18.24 points and $\eta = 5.456425$ under the stated transformation.

The cost sample spans ECI 96.62–146.24 and release years 2023–2025, across seven organizations. It measures cross-model training-cost levels. It does not identify the cost of marginally improving an incumbent, or the effort-cost schedule $c_0 a^{\gamma} e$ in equation (26). The slope-to-γ equality requires a technological and accounting relationship that is currently assumed.

Likewise, attributing the entire annual frontier movement to one unit of research combines compute expansion, algorithms, and inference/reasoning expenditure. The index-to-capability map determines its interpretation. For the alternative maintained map $a = \exp[(E - E_0)/s]$:

ECI points per unit of $\log a$: s	Implied cost exponent	Implied research coefficient
5	0.9269	39.2677
10	1.8538	5.4564
20	3.7076	1.5638

Changing s while retaining other normalized primitives is a different economic assumption. It is not a harmless unit change. Lemma 1 requires transforming thresholds, benefits, financing, and technologies consistently for invariance.

Essential revision: run each bridge separately; show the binding budgets and eligibility conditions; specify the reference index E_0 and the implied engineering thresholds in index units; and distinguish data moments from the maintained mapping into the economic model. Organization/recent-cohort cost sensitivity is useful, but cannot turn the association into a causal marginal-cost estimate. There is no identified real-world probability to recover by attaching additional decimal places to these scenarios.

9. Numerical audit and policy conclusions

9.1 Numerical claims checked

I checked the numerical claims in the prose, generated macros, all seven tables, figure-generation calculations, and the 42-row curvature/risk grid against supplied outputs. Appendix A below records the full empirical and sensitivity table ledger. Table 5 is notation; Table 6's numerical assumptions match the parameter file.

Selected headline checks are:

Claim	Reproduced value
Principal sample: observations / models / pairs	27,471 / 32 / 441
Principal quadratic estimate / SE	-0.100571 / 0.057058

Claim	Reproduced value
Approximate 95% interval	[-0.212404, 0.011262]
Distinct cost runs / slope / HC1 SE	18 / 1.853806 / 0.138364
Retrospective frontier change	18.24 ECI points
10-year open/open loss of control / extinction	3.295554% / 0%
25-year open/open loss of control / extinction	4.576073% / 4.358542%
25-year open/open welfare	774.896796
25-year ban/ban welfare	781.649157
25-year planner loss of control / extinction	0.105889% / <0.000001%
Bridge open/open research / delegation cycles	2 / 0
Bridge open/open welfare	527.752356

The manuscript's rounded values are consistent. Fradkin's cited ten-point/seven-percentage-point statement is also described accurately in its original source; I verified the statement, not his underlying analysis.

A small labeling repair is needed in experiments.run: three grid cells classified as "exposure without extinction by T" have positive probabilities below the 10^{-8} classification tolerance (approximately 2.39×10^{-15}, 1.38×10^{-11}, and 7.32×10^{-13}). Label these as numerically negligible by the horizon, rather than exactly zero. The distinction changes no displayed percentage or substantive policy comparison.

9.2 The government prisoner's dilemma is real within the specified continuations

Table 3's 25-year open/ban payoffs reproduce:

Jurisdiction 1 / 2	Open	Ban
Open	(387.428, 387.469)	(392.929, 385.444)
Ban	(385.444, 392.929)	(390.825, 390.825)

Each jurisdiction prefers open to ban against either action, while each prefers mutual ban to mutual open. The claimed prisoner's dilemma under $\sigma = 1.3$ is numerically valid. It is conditional on the selected continuation equilibria, horizon, action menus, and loss accounting. It is not established for every equilibrium selection or enforceability regime.

At ten years the selected government equilibrium is standard/standard; under the empirical bridges policies are payoff-equivalent. The full 25-year menu also has payoff-equivalent open/standard mixtures. These differences are correctly treated as scenario outcomes, not jurisdiction-specific predictions.

9.3 Which conclusions follow

The following conclusions are supported: commercial concavity alone does not locate the stopping point; one equilibrium can avoid delegation under the certificate; a parameter absent from an autonomous problem does not affect that problem at a fixed initial state; and eliminating acquisition eliminates the stipulated resource-conflict extinction channel.

Several stronger conclusions remain unsupported. The paper does not establish a general risk ordering of standards, bans, and planning, a quantitative effect of an actual international agreement, or a selection-robust commercial-curvature threshold. A ban yields zero modeled risk because delegation is assumed to be the only entry to the only extinction channel. This is a conditional implication, not independent evidence of practical policy sufficiency.

The supplied no-acquisition sensitivity is particularly informative: extinction becomes zero while loss of control rises from 4.576% to 5.050%. Removing a downstream harm changes upstream incentives. This illustrates why welfare, delegation, first loss, and extinction should be reported separately.

9.4 Small provenance repair

Location: acquire_data.arena. On a later invocation, the function fetches the current remote dataset revision even if it reuses an existing local Parquet without downloading it. It can then label old bytes with a fresh revision. Preserve original revision/retrieval metadata when reusing a cache, or require an explicit immutable revision. The normal frozen replication is unaffected, and I found no evidence that this snapshot is misattributed.

10. Editorial assessment and prioritized revisions

The manuscript has a coherent conditional question and a reproducible implementation. The current formal results and descriptive evidence nevertheless do not meet the stated journal ambition. Repairing the numerical descriptions is necessary, but technical tidiness alone would not justify a revise-and-resubmit recommendation at that standard.

The following order separates repairs available now from research extensions:

1. **Repair the computational-domain validation.** Cover feasible deviations and autonomous alternatives; retain path clipping as a diagnostic. Add the explicit false-stopping counterexample above as a regression check.
2. **Make the transition kernel and equilibrium search agree with the manuscript.** Choose shared-outcome or regime-specific projection explicitly; use full support enumeration or narrow the selection claim; separate acquisition cost from controlled safety resource use.
3. **Expose the actual stopping mechanisms.** Report financing, eligibility, and private-gain margins by state. Add smaller research inputs or justify minimum project size. Separate no research, no delegation, and no extinction by the horizon.
4. **Compute and develop Theorem 2.** Report the global delegated-deviation margin, its binding states, and economically meaningful primitive restrictions. A certificate using solved values should become a demonstrated analytical/computational contribution, not remain an unevaluated characterization.
5. **Decompose welfare and terminal effects.** Separate nonfatal economic losses from mortality, vary post-containment recovery and terminal coefficients individually, and report cumulative risks under one fixed long-horizon equilibrium.
6. **Strengthen the empirical audit without pretending to identify what the data cannot identify.** Add attrition/model support, model-node inference sensitivity, mapping/budget provenance, and the common-sample nonmonotonicity. Preserve the descriptive label and absence of a monetary σ estimate.
7. **Unbundle the calibration bridges.** Run each alone and show how capability scaling, unit project size, finance, and launch eligibility determine the resulting path. Fix cached-data provenance.

8. **Make the contribution decision explicit.** A general-interest submission needs a new economically informative characterization or materially stronger evidence. Additional arbitrary scenario sweeps will not substitute for either.

Multiple uncontrolled systems, relocation, hidden actions, belief heterogeneity, richer resource dynamics, and an infinite-horizon model are possible extensions. They are not prerequisites for correcting this draft, and the revision should not promise them merely to enlarge the paper. Nor should it promise an identified extinction probability.

Editorial conclusion: reject the present draft at the requested standard. The central claims that survive this audit are conditional possibility and stopping statements. A stronger paper would explain, through primitives or evidence, when those conditions bind and how enforceable constraints change them.

Appendix A. Complete table ledger

All entries below match the supplied outputs and regenerated manuscript tables to displayed precision. Additional diagnostics elsewhere in the report are referee calculations.

A.1 Table 1: preferences

Specification	N	Models	Linear (SE)	Quadratic (SE)
Linear ECI	27,471	32	0.483 (0.037)	–
Quadratic ECI	27,471	32	0.419 (0.053)	-0.101 (0.057)
Ties one-half	33,318	32	0.347 (0.044)	-0.087 (0.047)
Qualified aliases	52,764	41	0.515 (0.049)	-0.016 (0.053)
GPQA common	9,183	16	0.259 (0.035)	0.050 (0.020)
ECI common	9,183	16	0.027 (0.146)	-0.339 (0.143)
Coding	7,481	32	0.474 (0.062)	-0.187 (0.071)
Other	19,990	32	0.395 (0.058)	-0.075 (0.062)
English	15,127	32	0.379 (0.057)	-0.100 (0.062)

A.2 Table 2: policies

Specification	Policy	First loss (%)	Extinction (%)	Welfare
10 years	open/open	3.296	0.000	509.466
10 years	standard/open	3.291	0.000	509.467
10 years	standard/standard	3.286	0.000	509.468
10 years	ban/ban	0.000	0.000	503.953
10 years	Planner	3.194	0.000	509.499
25 years	open/open	4.576	4.359	774.897
25 years	standard/open	4.576	4.359	774.897
25 years	standard/standard	4.576	4.359	774.897
25 years	ban/ban	0.000	0.000	781.649
25 years	Planner	0.106	0.000	781.671
Bridges, 10 years	open/open	0.000	0.000	527.752
Bridges, 10 years	standard/open	0.000	0.000	527.752

Specification	Policy	First loss (%)	Extinction (%)	Welfare
Bridges, 10 years	standard/standard	0.000	0.000	527.752
Bridges, 10 years	ban/ban	0.000	0.000	527.752
Bridges, 10 years	Planner	0.000	0.000	527.752

A.3 Table 4: family exclusions

Excluded family	N	Quadratic	SE
Anthropic	15,355	-0.112	0.058
DeepSeek	23,385	-0.044	0.060
Google	19,331	-0.131	0.063
OpenAI	16,337	-0.338	0.099
Qwen	21,166	-0.102	0.065
grok	26,932	-0.120	0.060
kimi	26,613	-0.099	0.058
llama	23,015	-0.028	0.068
mistral	23,681	-0.053	0.062

A.4 Table 7: sensitivity

Specification	First loss (%)	Extinction (%)
5 years	4.424	0.000
25 years	4.576	4.359
No acquisition, 25 years	5.050	0.000
Full common loss shock, 25 years	2.719	2.594
High-risk pure selection, 25 years	4.800	4.581
Half capability mesh, 25 years	4.561	4.353
Double capability ceiling, 25 years	4.576	4.358
Zero terminal values, 25 years	4.243	4.025
Long terminal values, 25 years	5.646	4.108
No containment, 25 years	4.574	4.574
No positional prize, 25 years	3.211	3.001
Low research feedback, 25 years	4.357	3.871
High research feedback, 25 years	3.786	3.692
Bridge: half cost slope	0.810	0.000
Bridge: double finance	0.000	0.000
Bridge: ready to delegate	0.810	0.000
Bridge: higher market scale	0.000	0.000

Every displayed boundary entry in Table 7 is 0.0000%; every recorded selected-path boundary value in the 42-row risk grid is zero. As Section 3 explains, this is not a certificate against off-path truncation.

A.5 Other numerical text

The raw Arena dates run from April 17 to July 24, 2025. There are 270 ECI entries and 3,628 Epoch model entries. The principal sample has 441 unordered pairs; the common sample has 120. Three identified shared-training derivatives are excluded before retaining 18 cost runs. The reference parameters in Table 6 agree with the parameter file. The static counterexample and prisoner's-dilemma payoffs are checked in Sections 1 and 9. Both figure-generating scripts use the supplied estimates/scenarios, and their plotted probability units and specified support restrictions are correct.

Appendix B. Evidence and reproducibility record

The primary reviewed materials are the uploaded paper.pdf, bundle.txt, and repository-snapshot.zip, cross-checked against the exact public GitHub commit. All manuscript equation and table references in this report use that frozen PDF's numbering.

The review executed the repository unchanged in an isolated copy. Targeted checks altered functions only in the audit process and did not edit the frozen project. These checks covered the false-stopping ceiling example; full support enumeration for the 25-year reference; a common projected research transition; addition of the autonomous budget-boundary effort; and descriptive dyadic/cost/matching diagnostics.

The successful full replication verifies execution and the supplied hash manifest. It does not independently verify every third-party model/version claim, re-estimate ECI, audit the providers' original training-cost accounts, or identify the scenario assumptions. The original reference-paper cache is excluded from the supplied archive; separately accessed sources are listed below.

Primary external sources consulted

- Jones, Charles I. (2024), *The AI Dilemma: Growth versus Existential Risk*, American Economic Review: Insights 6(4), 575–590. Published paper on the author's Stanford page.
- Growiec, Jakub, and Klaus Prettner (2026), *The Economics of p(doom): Scenarios of Existential Risk and Economic Growth in the Age of Transformative AI*, April 24 version, arXiv:2503.07341v2. Accessible PDF.
- Bueno de Mesquita, Ethan, Wioletta Dziuda, and Mattias Polborn (2026), *The AGI Race and Existential Risk*, BFI Working Paper 2026-71, May version. BFI paper. This accessible copy is dated May 25; it is used for the literature comparison, not asserted to be byte-identical to the author's excluded May source cache.
- Cunningham, Tom, et al. (2026), *The Economics of Recursive Self-Improvement*, arXiv:2609.15802v1. ArXiv record and available HTML.
- Arena Team, *A Deep Dive into Recent Arena Data*. Source documentation.
- Epoch AI, *ECI documentation: Overview*. Source methodology overview.
- Fradkin, Andrey (2026), *Measuring Diminishing Returns to LLM Intelligence*, July 27. Original preliminary analysis.
- Frozen public project: GitHub commit 5a1d9a4. Tree membership and blob hashes were accessed through GitHub's public API for this exact commit.